

RELAY Inference Algorithm with Gaussian Hidden-State Noise

Technical specification of the stochastic self-conditioning inference procedure

1. State Initialization

For a Sudoku of length $L = 81$, define the state at inference iteration k as

$$\begin{aligned} s_k &= (x^{(k)}, h^{(k)}), \\ x^{(k)} &\text{ in } \mathbb{V}^{81}, \quad h^{(k)} \text{ in } \mathbb{R}^{(81 \times 384)}. \end{aligned}$$

At clue positions, $x^{(0)}$ contains the given digit; at empty positions it contains [MASK].

Initialize $h^{(0)} = 0$. Let $c_i = 1$ denote an immutable clue position.

2. RELAY Forward Pass

At iteration k , RELAY combines token embeddings with the relayed hidden state:

$$\begin{aligned} u^{(k)} &= \text{Embed_theta}(x^{(k)}) + \text{LN_relay}(h^{(k)}), \\ z^{(k)} &= \text{Transformer_theta}(u^{(k)}). \end{aligned}$$

The same forward pass produces token logits and the Gaussian hidden-policy mean:

$$\begin{aligned} \Lambda^{(k)} &= \text{LMHead_theta}(\text{LN_out}(z^{(k)})), \\ \mu^{(k)} &= z^{(k)}. \end{aligned}$$

$\Lambda^{(k)}$ defines token probabilities; $\mu^{(k)}$ defines the mean of the hidden Gaussian policy.

3. Token Probabilities

Full-vocabulary token probabilities use token temperature $\tau_{\text{tok}} > 0$:

$$P_{[i,v]}^{(k)} = \frac{\exp(\Lambda_{[i,v]}^{(k)} / \tau_{\text{tok}})}{\sum_u \exp(\Lambda_{[i,u]}^{(k)} / \tau_{\text{tok}})}.$$

For every selected position i , sample

$$\hat{x}_i^{(k)} \sim \text{Categorical}(P_{[i,:]}^{(k)}).$$

An equivalent Gumbel-Max implementation is

$$\begin{aligned} \hat{x}_i^{(k)} &= \arg\max_v [\Lambda_{[i,v]}^{(k)} / \tau_{\text{tok}} + g_{[i,v]}], \\ g_{[i,v]} &\sim \text{Gumbel}(\theta, 1). \end{aligned}$$

4. Confidence-Based Position Selection

The currently masked and mutable positions are

$$M^{(k)} = \{i : \hat{x}_i^{(k)} = [\text{MASK}] \text{ and } c_i = 0\}.$$

Confidence uses a separate temperature T_{conf} :

$$\begin{aligned} P_{\text{conf}[i,v]}^{(k)} &= \text{softmax}(\Lambda_{[i,:]}^{(k)} / T_{\text{conf}})_v, \\ a_i^{(k)} &= \max_v P_{\text{conf}[i,v]}^{(k)}. \end{aligned}$$

Sort the mutable masked positions by decreasing confidence:

$$a_{\pi_1}^{(k)} \geq a_{\pi_2}^{(k)} \geq \dots \geq a_{\pi_n}^{(k)}, \quad n = |M^{(k)}|.$$

Define cumulative uncertainty and the initial selected set:

$$\begin{aligned} C_j^{(k)} &= \sum_{(r=1)}^j (1 - a_{\pi_r}^{(k)}), \\ U_{\text{tilde}}^{(k)} &= \{\pi_j : C_j^{(k)} < \eta\}. \end{aligned}$$

The final selected set is

$$\begin{aligned}
U^k &= U_{\text{tilde}}^k, & \text{if } U_{\text{tilde}}^k \text{ is nonempty;} \\
&= \{p_{i_1}\}, & \text{if } U_{\text{tilde}}^k \text{ is empty and } M^k \text{ is nonempty;} \\
&= \text{empty set}, & \text{if } M^k \text{ is empty.}
\end{aligned}$$

The fallback guarantees that at least one position is selected while unresolved masks remain.

5. Token-State Transition

Only selected positions are updated:

$$\begin{aligned}
x_i^{k+1} &= \hat{x}_i^k, & \text{if } i \text{ is in } U^k; \\
&= x_i^k, & \text{otherwise.}
\end{aligned}$$

Clue positions are never modified. This is inference: sampled tokens are written into the next state, and no teacher forcing is used.

6. Gaussian Hidden-State Transition

The hidden policy is

$$q_{\theta}(h^{k+1} \mid s_k) = \text{Normal}(h^{k+1}; \mu^k, \sigma^2 I).$$

Sample independent Gaussian noise and update the relayed hidden state:

$$\begin{aligned}
\epsilon^k &\sim \text{Normal}(0, I), \\
h^{k+1} &= \mu^k + \sigma \epsilon^k \\
&= z^k + \sigma \epsilon^k.
\end{aligned}$$

The sampled hidden state is fed into the next forward pass:

$$\begin{aligned}
u^{k+1} &= \text{Embed}_{\theta}(x^{k+1}) \\
&+ \text{LN}_{\text{relay}}(z^k + \sigma \epsilon^k).
\end{aligned}$$

When $\sigma = 0$, hidden relaying is deterministic. When $\sigma > 0$, self-conditioning is stochastic.

7. Complete State Transition

One complete inference iteration is

$$\begin{aligned}
s_k &= (x^k, h^k); \\
(\Lambda^k, \mu^k) &= f_{\theta}(x^k, h^k); \\
U^k &= \text{ConfidenceSelect}(\Lambda^k, x^k, c; T_{\text{conf}}, \eta); \\
\hat{x}_i^k &\sim \text{Categorical}(\text{softmax}(\Lambda_{[i,:]}^k / \tau_{\text{tok}})); \\
x_i^{k+1} &= \hat{x}_i^k \text{ for } i \text{ in } U^k, \text{ and } x_i^k \text{ otherwise;} \\
h^{k+1} &= \mu^k + \sigma \epsilon^k; \\
s_{k+1} &= (x^{k+1}, h^{k+1}).
\end{aligned}$$

8. Termination and Number of Forward Evaluations

Inference terminates when no mutable masks remain, M^{k+1} is empty, or when the maximum rollout length K_{max} is reached. At the final allowed iteration, all remaining mutable mask positions are forced to be selected.

NFE = number of executed inference iterations.

Each iteration requires one RELAY forward evaluation.

9. Current Experimental Configuration

$T_{\text{conf}} = 2$, $\tau_{\text{tok}} = 0.3$, $\eta = 0.15$.

- $\sigma_{\text{eval}} = 0$: deterministic hidden relaying.

- `sigma_eval = 1`: Gaussian hidden-state relaying.